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### ApisQueen U2 MINI+45A ESC

# ApisQueen U2 Mini Underwater Thruster with 45A ESC User Manual

Model: U2 MINI+45A ESC

## Introduction

This manual provides essential instructions for the safe and efficient operation, setup, and maintenance of your ApisQueen U2 Mini Underwater Thruster with 45A ESC. Please read this manual thoroughly before use to ensure optimal performance and longevity of your device.

## Safety Information

- Always ensure proper electrical connections before powering on the thruster. Incorrect wiring can lead to damage or injury.
- Do not connect the thruster directly to a battery without the Electronic Speed Controller (ESC). The ESC is required for proper voltage regulation and control.
- Keep hands and loose objects clear of the propeller when the thruster is powered to prevent injury.
- Operate the thruster in a well-ventilated area, especially during testing, to avoid overheating components.
- Ensure the battery used is compatible with the thruster's voltage requirements (12-16V, 3-4S LiPo).
- After use, disconnect the battery from the thruster and ESC.

## Package Contents

Verify that all items are present in your package:

- U2 mini CCW thruster x1
- 45A ESC x1

## Specifications

| Feature | Detail |
|---|---|
| Brand | ApisQueen |

| | |
|---|---|
| Model Name | U2 MINI(CCW)+45A ESC |
| Voltage | 12-16 Volts (3-4S LiPo) |
| Max Current | 8A |
| Max Power | 130W |
| Thrust | Up to 2.87 lbs (1.3 Kg) |
| Motor KV | 500KV |
| ESC Type | 45A Bi-directional Waterproof ESC, supports OneShot 125 |
| Material | Marine-grade composites or alloys |
| Cable Length | >900mm |
| Weight | 210g |

*Note: The thruster cannot be connected directly to the battery; an ESC is required.*




Image: ApisQueen U2 Mini Underwater Thruster with key specifications including thrust, voltage, current, power, size, cable length, and weight.

| APISQUEEN U2 Mini UNDERWATER THRUSTER | | | |
|---|---|---|---|
| Forward | | | |
| Voltage ( V ) | Throttle | Current ( A ) | Thrust ( Kg) |
| 10 | 100% | 4 | 0.70 |
| 12 | 100% | 5 | 0.90 |
| 14 | 100% | 7.0 | 1.10 |
| 16 | 100% | 8 | 1.30 |
| Reversal | | | |
| Voltage ( V ) | Throttle | Current ( A ) | Thrust ( Kg) |
| 10 | 100% | 4 | 0.60 |
| 12 | 100% | 5 | 0.80 |
| 14 | 100% | 7 | 1.00 |
| 16 | 100% | 8 | 1.20 |

**Note: The thruster can not be connected directly to the battery, you need to use ESC.**

Image: Performance table detailing forward and reversal thrust (Kg) at various voltages (V) and 100% throttle, along with corresponding current (A).

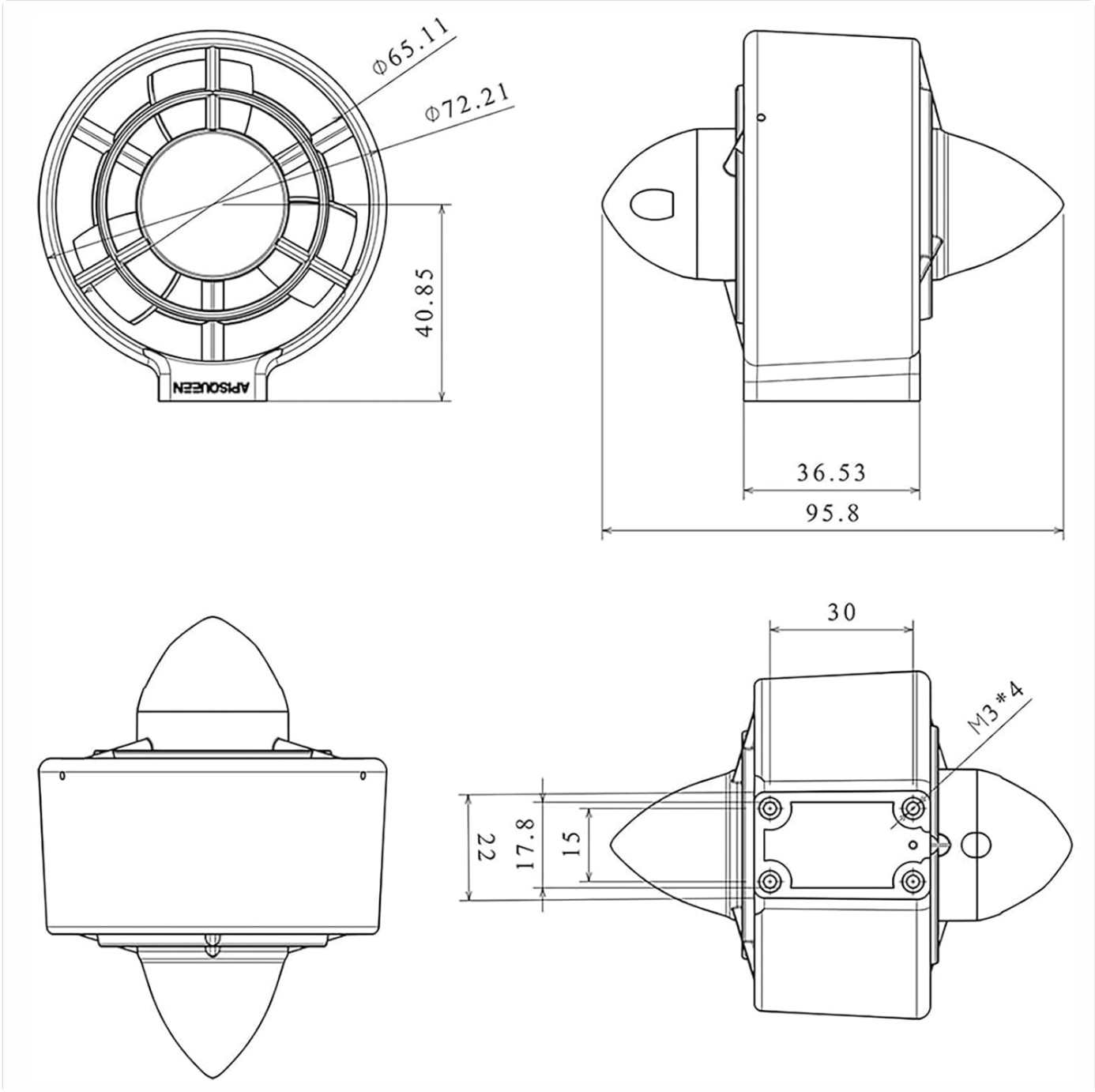



Image: Technical drawing showing the dimensions of the ApisQueen U2 Mini Underwater Thruster, including diameter, length, and mounting hole details.

## Setup Instructions

Follow these steps to correctly set up your U2 Mini Thruster with the 45A ESC and a remote control system.

### Wired Control Method

1. Ensure the remote control is turned on.
2. Insert the UBEC into any signal receiver frame, excluding the first frame.
3. Insert the thruster signal cable into the first frame of the signal receiver.
4. Plug the UBEC into the battery.
5. Connect the thruster to the battery. Upon connection, the thruster will emit a "bi-bi-bi" sound, indicating it has entered the working state.
6. If the water outlet direction is incorrect, change any two of the three motor wires connected to the ESC.

7. If the ESC does not have a BEC 5V output and cannot power the receiver, you will need to provide additional power to the remote control receiver. If the signal line set has only 2 lines, it indicates the ESC does not have a BEC and requires additional power supply.
8. For wired control, connect the signal receiver to a PWM Pulse Width Modulator. The metal side of the connector faces upward (small white, red, and black wires are connected to S+, -, + of the OUT port respectively).
9. Initially, turn the knob on the PWM modulator to the middle position (1.5ms).

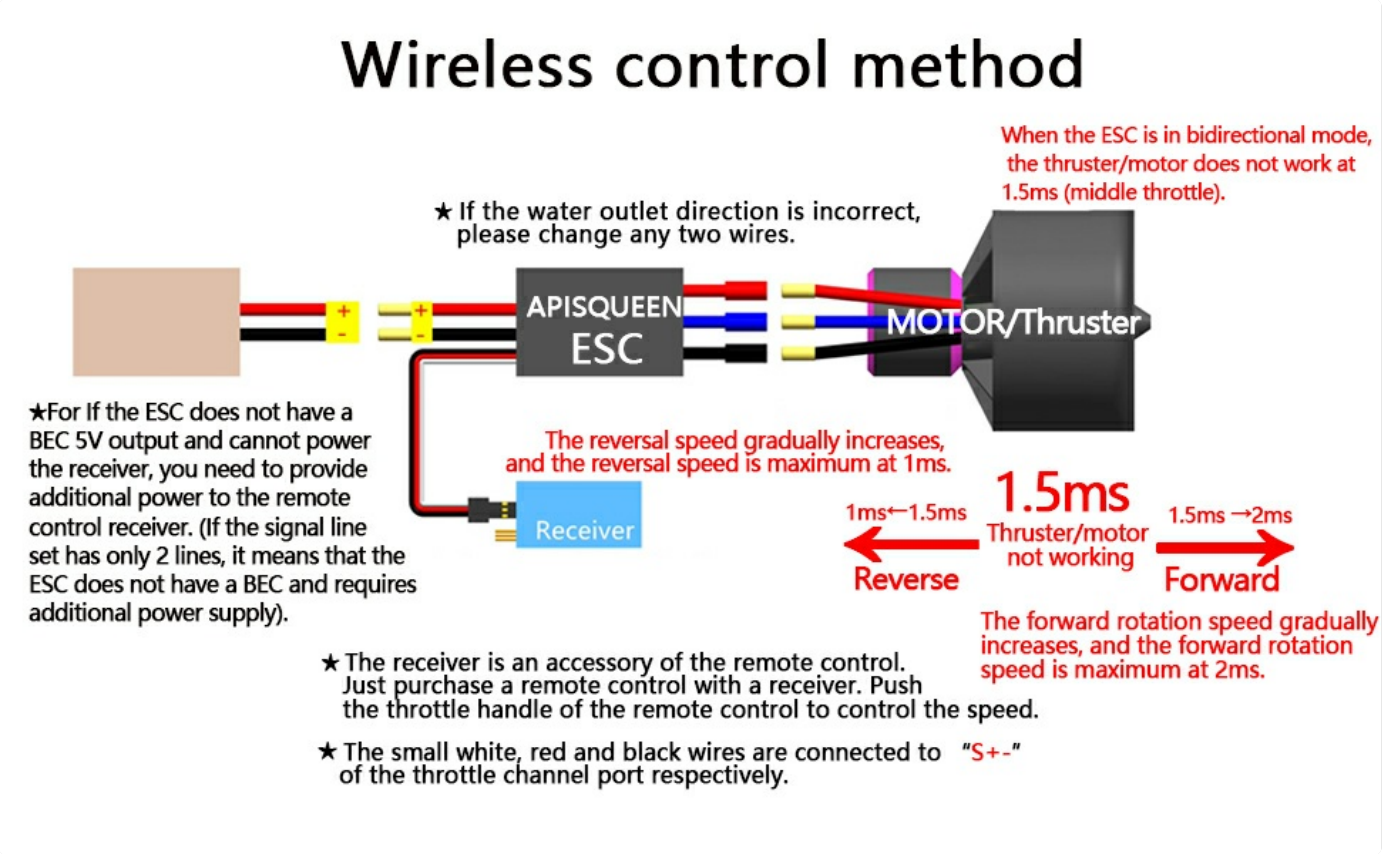



Image: Wiring diagram illustrating the connection of the ApisQueen ESC to the motor/thruster, a signal receiver, and a PWM modulator for wired control. It also shows the throttle signal ranges for forward and reverse operation.

## Wireless Control Method

1. Ensure the remote control is turned on.
2. Insert the UBEC into any signal receiver frame, excluding the first frame.
3. Insert the thruster signal cable into the first frame of the signal receiver.
4. Plug the UBEC into the battery.
5. Connect the thruster to the battery. Upon connection, the thruster will emit a "bi-bi-bi" sound, indicating it has entered the working state.
6. If the water outlet direction is incorrect, change any two of the three motor wires connected to the ESC.
7. If the ESC does not have a BEC 5V output and cannot power the receiver, you will need to provide additional power to the remote control receiver. If the signal line set has only 2 lines, it indicates the ESC does not have a BEC and requires additional power supply.
8. For wireless control, the receiver is an accessory of the remote control. Connect the small white, red, and black wires of the throttle channel port respectively. Push the throttle handle of the remote control to control the speed.

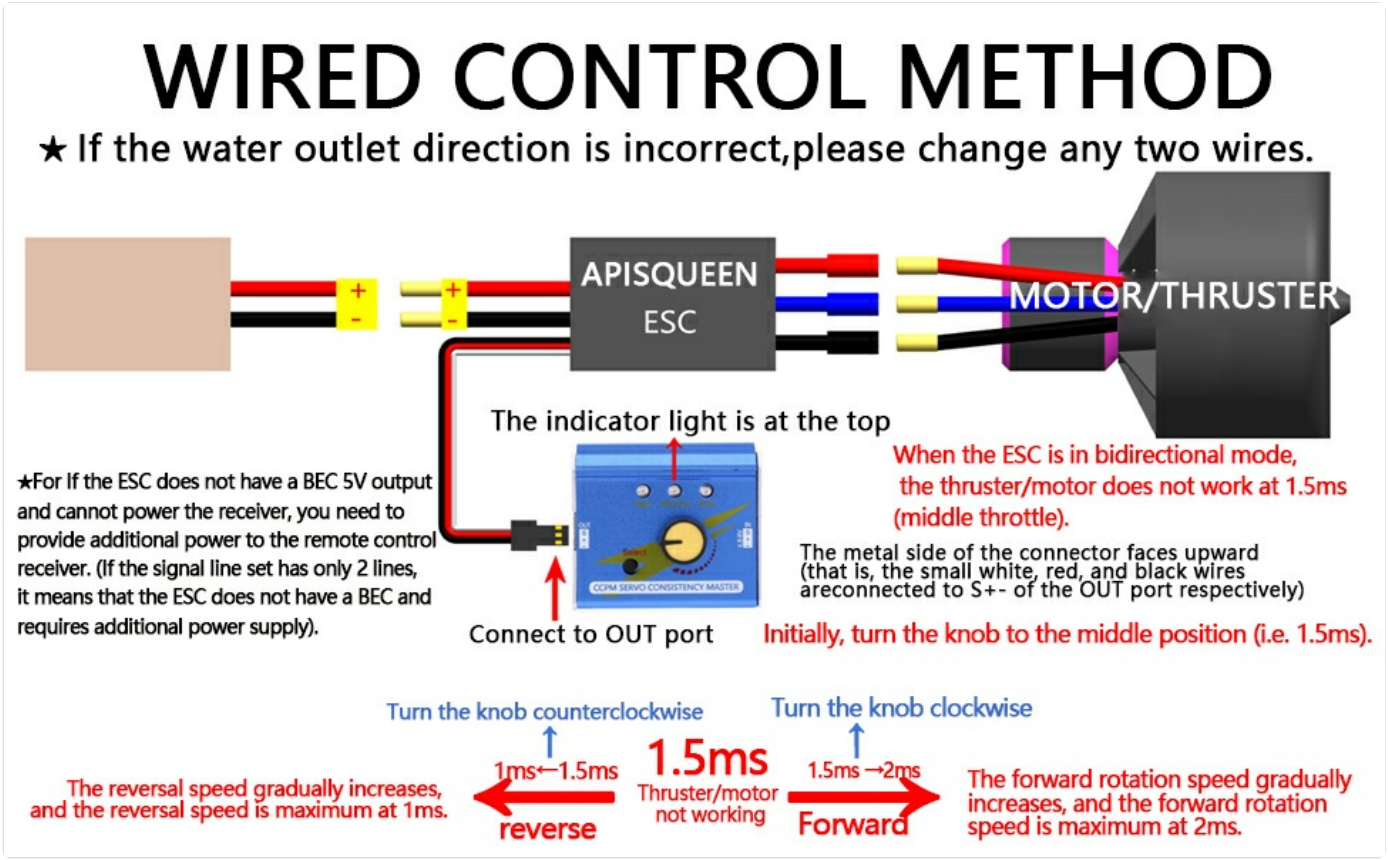



Image: Wiring diagram illustrating the connection of the ApisQueen ESC to the motor/thruster, a signal receiver, and a remote control for wireless operation. It also shows the throttle signal ranges for forward and reverse operation.

## Visual Setup Guide (Video)

Your browser does not support the video tag.

Video: A step-by-step guide demonstrating the connection process for the ApisQueen U2 Mini Thruster, including connecting the UBEC to the signal receiver, the thruster signal cable, and the battery. It also shows testing the forward and backward thrust.

## Operating Instructions

Once the thruster system is correctly set up and powered, you can control its operation using your remote control or PWM modulator.

- **Forward Thrust:** Increase the throttle signal from the neutral position (1.5ms) towards 2ms. The thruster will begin to rotate, generating forward thrust.
- **Reverse Thrust:** Decrease the throttle signal from the neutral position (1.5ms) towards 1ms. The thruster will rotate in the opposite direction, generating reverse thrust.
- **Neutral Position:** At 1.5ms throttle signal, the thruster/motor will not operate. This is the idle or stop position.
- Always ensure the thruster is submerged in water before operating at high speeds to prevent damage from cavitation or overheating.

## Maintenance

Regular maintenance ensures the longevity and reliable performance of your U2 Mini Underwater Thruster.

- **Rinsing:** After each use in saltwater, thoroughly rinse the thruster and ESC with fresh water to prevent salt buildup and corrosion.
- **Inspection:** Periodically inspect the propeller for any debris, damage, or entanglement. Remove any foreign objects

carefully.

- **Cable Check:** Examine all cables and connectors for signs of wear, cuts, or corrosion. Replace damaged components immediately.
- **Storage:** Store the thruster in a dry, cool place away from direct sunlight and extreme temperatures.

## Troubleshooting

If you encounter issues with your ApisQueen U2 Mini Thruster, refer to the following common problems and solutions:

| Problem | Possible Cause | Solution |
|---|---|---|
| Thruster does not operate or respond to control. | ◦ Incorrect wiring.<br>◦ Battery not connected or low charge.<br>◦ Remote control not powered on or not paired.<br>◦ ESC not receiving signal. | ◦ Verify all connections according to the setup diagrams.<br>◦ Check battery charge and connection.<br>◦ Ensure remote control is on and properly paired with the receiver.<br>◦ Confirm signal cable is correctly inserted into the receiver. |
| Thruster rotates in the wrong direction. | Motor wire polarity is incorrect. | Swap any two of the three motor wires connecting the thruster to the ESC. |
| Thruster makes unusual noises or vibrates excessively. | ◦ Debris in the propeller.<br>◦ Damaged propeller.<br>◦ Loose mounting. | ◦ Inspect and clear the propeller of any debris.<br>◦ Check for propeller damage and replace if necessary.<br>◦ Ensure the thruster is securely mounted. |
| ESC or motor gets excessively hot. | ◦ Overload or prolonged high-power operation.<br>◦ Insufficient cooling (not submerged).<br>◦ Faulty component. | ◦ Reduce power output or operating time.<br>◦ Ensure thruster is fully submerged during operation.<br>◦ Disconnect power immediately and contact support if overheating persists. |

## Warranty and Support

ApisQueen products are designed for quality and performance. For warranty information, technical support, or service inquiries, please refer to the contact information provided with your purchase or visit the official ApisQueen website. Keep your purchase receipt as proof of purchase for warranty claims.

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